

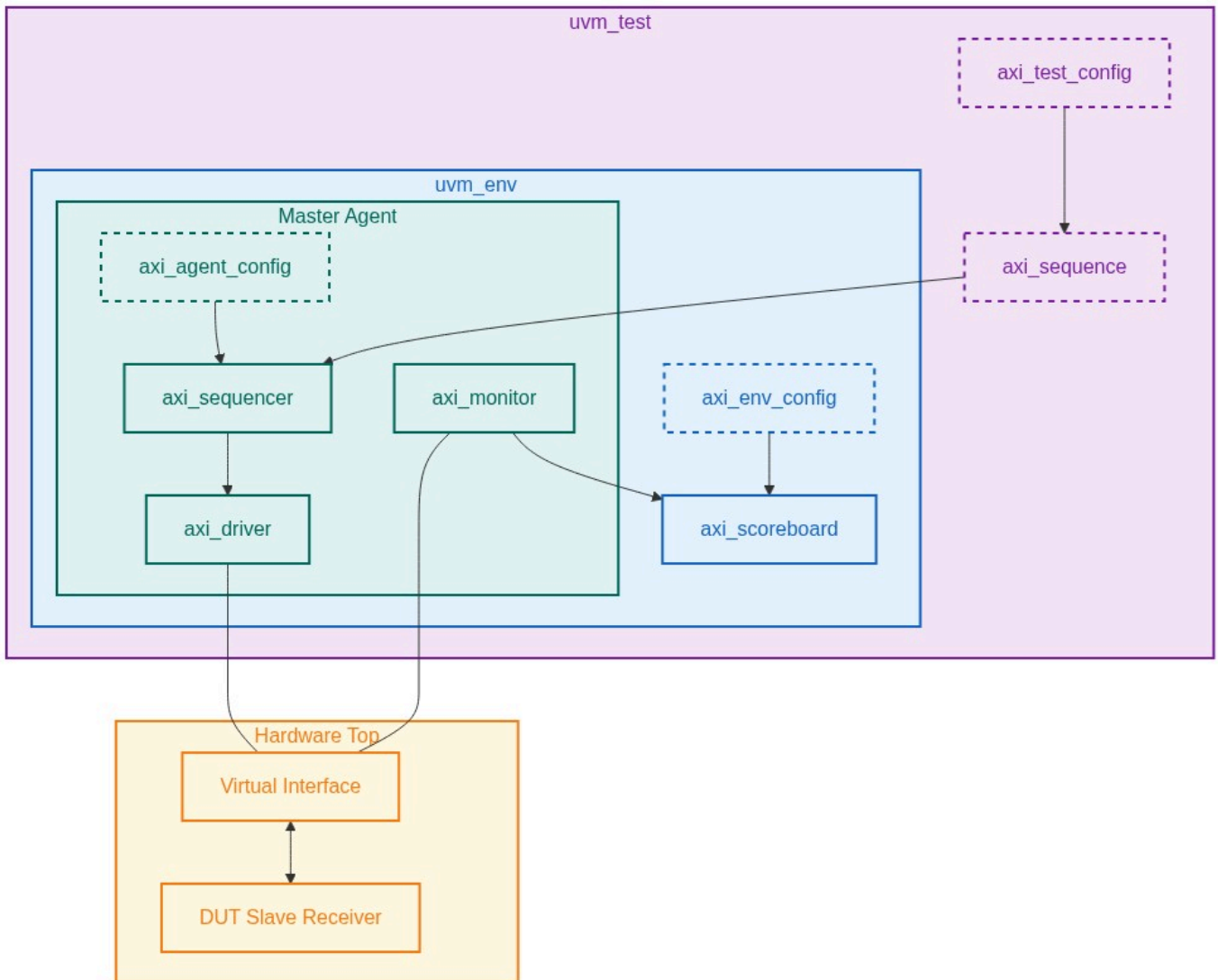
Verification Plan Report

Project: AXI-Stream Master VIP Verification Plan

Version: 8.0

Verification Role: Master

1. UVM Environment Architecture



2. Protocol Configurations

Feature	Configuration Name	Impact
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Handshake Protocol	TVALID_TREADY_Async_Handshake	Functionality impact: Forces pipeline stalling and dictates stability bounds for the entire payload vector. Performance impact: dictates maximum theoretical throughput in scenarios involving combinatorial TREADY backpressure.
Packet Framing	TLAST_Packet_Framing_Support	Functionality impact: Requires end-to-end exact reconstruction of packet boundary assertions to govern data routing layer and destination buffers. Performance impact: dictates fragmentation penalties and dynamic arbitration thresholds.
Byte-Level Stream Semantics	TKEEP_TSTRB_Qualifier-Encoding	Functionality impact: necessitates robust reconstruction algorithms in the scoreboard to extract physical data while discarding positional or null bytes. Performance impact: alters actual byte-consumption rate compared to clock-cycle metrics.
Reset Sequencing	ARESETn_Mid_Transfer_Isolation	Functionality impact: compels the testing environment to trigger state-machine destruction mid-transfer. Verification constraint: requires the VIP driver to safely discard tracking state and the monitor to correctly sequence clean IDLE resets asynchronously.
Wakeup Signaling (AXI5)	TWAKEUP_Pre_Transfer_Signaling	Functionality impact: introduces an external deadlock dependency if TWAKEUP assertion lags TVALID assertion. Performance impact: latency models must explicitly adjust for TWAKEUP lead times before TREADY is physically accessible.
Stream Identification	TID_TDEST_Interleaving_Routing	Functionality impact: forces the scoreboard multiplexer logic to concurrently maintain dynamically allocated data streams to track ordering compliance. Performance impact: tests the DUT buffer saturation across varying interleave depths.
Parity Protection (AXI5)	Odd_Parity_Check_Generation	Functionality impact: requires single-bit error injection capabilities within the VIP transmission chain to exercise the DUT standard error logic. Verification constraint: the monitor must passively track hardware parity matrices parallel to the native data payload across all TKEEP permutations.

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Continuous Packets Mode	Continuous_Packet_Constraint_Mode	Functionality impact: strictly constrains the randomization matrices of the VIP to avoid null-byte and inter-packet gaps. Performance impact: maximizes arbitration efficiency and validates peak continuous payload saturation without recovery cycles.
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3. Test Requirements

Index	Name	Description
REQ_MST_01	TVALID Stability and Handshake Integrity	The AXI-Stream handshake is predicated on the invariant that once a Transmitter asserts TVALID, it must hold TVALID HIGH and maintain all associated payload signals (TDATA, TSTRB, TKEEP, TLAST, TID, TDEST, TUSER) in a stable state until the handshake completes via coincident TREADY assertion. Critically, the Transmitter (VIP Master) is prohibited from conditioning TVALID assertion on a prior TREADY observation ? doing so would introduce a combinatorial dependency that risks deadlock. Verification Intent: The VIP Master driver must assert TVALID and hold all sideband signals with strict stability until TREADY is observed. The monitor will assert on mid-handshake mutations. We will inject TVALID drops via VIP configurations to characterize DUT responses. Under no circumstance should the DUT require non-standard behaviors; it should trigger its native state exception handling.

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REQ_MST_02	Packet Boundary Generation via TLAST Assertion	TLAST is the exclusive packet-boundary indicator generated by the Transmitter (VIP Master). The number of TLAST assertions must be preserved end-to-end between Transmitter and Receiver without additional assertions (false boundaries) or suppressions (missed boundaries). Verification Intent: VIP sequences must systematically cover variable-length transfers (1 to 256 beats) and continuous consecutive packets with zero gap. The scoreboard will parse standard TLAST assertions and assert data grouping correctness, entirely driven by configurable VIP framing intervals.
REQ_MST_03	Byte Qualifier Conformance - TKEEP and TSTRB Encoding	TKEEP governs byte transport eligibility. The combination TKEEP=LOW/TSTRB=HIGH is architecturally reserved and constitutes a protocol violation. Verification Intent: VIP sequences must systematically vary TKEEP and TSTRB across all byte lanes, including driving reserved configurations as negative stimulus. The DUT is assumed to operate securely; the environment will observe its native error responses when standard illegal byte masks are forced from the VIP.
REQ_MST_04	Clock and Reset Protocol - TVALID Deassertion During ARESETn	During ARESETn assertion, TVALID must be driven LOW. Premature TVALID assertion post-reset constitutes a protocol violation that may corrupt the DUT Slave's initialization state. Verification Intent: The VIP reset-sequence handles overriding the protocol state. Reset events will be driven asynchronously by the VIP mid-packet and mid-handshake. The DUT must correctly flush standard buffers according to spec without requiring custom rescue stimulus from the agent.

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REQ_MST_05	AXI5 Parity Generation Across All Check Signals	The AXI5-Stream parity protection scheme employs odd parity across all interface signals. TDATACHK must be driven for all byte lanes unconditionally (even null lanes). Verification Intent: The VIP will continuously calculate and drive proper odd parity matrices. Test case scenarios will actively corrupt single bits on the VIP transmitter driver to stimulate the DUT's standard error interrupts. The environment will not require the DUT to support custom diagnostic overrides.
REQ_MST_06	Multi-Stream Interleaving Integrity	Inter-stream interleaving at per-transfer granularity is permitted. The VIP Master must maintain independent stream contexts for each active TID value and must not reorder beats within a shared stream context. Verification Intent: The VIP will leverage variable-rate sequence arbitration to interleave up to four concurrent streams on a beat-granularity. The scoreboard will reconstruct data segments without modifying or restricting DUT routing thresholds.

4. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_MST_001	TVALID-First Handshake: VIP Master sequence asserts TVALID with stable TDATA. The VIP delays subsequent action while DUT Slave holds TREADY randomly. The VIP monitor confirms payload stability across the stalling boundary.	REQ_MST_01
TC_MST_002	Negative Parameterized Drop: VIP is instructed to deliberately withdraw TVALID before TREADY assertion. The standard DUT behavior is observed and verified against spec compliance.	REQ_MST_01

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TC_MST_003	Back-to-Back Packet Sequence: VIP sequence drives a continuous payload where TLAST=1 is immediately followed by a new packet with TVALID=1 in the very next cycle without gap. The scoreboard verifies standard DUT routing extraction.	REQ_MST_02
TC_MST_004	Sparse Stream Position Sequence: VIP drives randomized TKEEP variations covering fully null packages, position bytes, and mixed data streams. The scoreboard validates physical byte extraction.	REQ_MST_03
TC_MST_005	Reserved Encoding Injection: VIP driver forces sequence TKEEP=0/TSTRB=1 locally to trigger a negative constraint. Standard DUT reporting thresholds (if implemented) are captured.	REQ_MST_03
TC_MST_006	Mid-Packet Asynchronous Reset: While VIP holds TVALID=1/TREADY=0, the VIP synchronously drives ARESETn LOW. Confirm DUT reliably drops pipeline and recovers on its own.	REQ_MST_04
TC_MST_007	AXI5 Parity Single-Bit Inversion: VIP driver mathematically calculates correct parity, then inverts bit 1 of TDATACHK array. The monitor logs standard DUT protocol violation responses.	REQ_MST_05
TC_MST_008	High-Depth Multi-Sequence Interleave: Four parallel VIP sequences drive disjoint TID packets into a shared driver. The VIP arbitrates tightly, generating a deeply interleaved stream. The scoreboard tests DUT tracking stability.	REQ_MST_06

5. Functional Coverage Groups

Group Name	Description
cg_handshake_ordering	Standard cross-coverage of TVALID/TREADY timing relations and VIP-induced wait-states.
cg_packet_boundary_dimensions	Coverage bins for packet sizes dynamically constrained by the VIP sequences (single, multi, continuous).
cg_byte_lane_permutations	Coverage group mapping the TKEEP driver metrics against legal and illegal VIP combinations.
cg_reset_window_injections	Vector tracking exactly where in the transfer phase the VIP interrupted the DUT via ARESETn.
cg_parity_failure_positions	Tracks individual bits inverted by the VIP driver to ensure the DUT parity path is fully exercised.
cg_stream_interleave_depths	Records max concurrent sequence IDs handled cleanly by the standard DUT architecture.

6. Interface Signals

Signal Name	Direction	Width	Description
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ACLK	Input (to VIP Master and DUT Slave)	1	Global synchronization clock.
ARESETn	Input (to VIP Master and DUT Slave)	1	Active-LOW global reset.
TVALID	Output (VIP Master / Transmitter)	1	Driven by the VIP Master. Indicates a valid transfer payload is present.
TREADY	Input (from DUT Slave / Receiver)	1	Driven by the DUT Slave to signal readiness to accept a transfer.
TDATA	Output (VIP Master / Transmitter)	TDATA_WIDTH	Primary data payload.
TSTRB	Output (VIP Master / Transmitter)	TDATA_WIDTH / 8	Per-byte data qualifier.
TKEEP	Output (VIP Master / Transmitter)	TDATA_WIDTH / 8	Per-byte position qualifier.
TLAST	Output (VIP Master / Transmitter)	1	Packet-boundary indicator.
TID	Output (VIP Master / Transmitter)	TID_WIDTH	Stream identifier.
TDEST	Output (VIP Master / Transmitter)	TDEST_WIDTH	Routing identifier.
TVALIDCHK	Output (VIP Master / Transmitter, AXI5-Stream only)	1	Odd parity check bit for TVALID.

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TDATACHK	Output (VIP Master Transmitter, AXI5-Stream only)	TDATA_WIDTH / 8	Odd parity check per byte lane of TDATA.
TLASTCHK	Output (VIP Master Transmitter, AXI5-Stream only)	1	Odd parity check bit for TLAST.
TREADYCHK	Input (from DUT Slave Receiver, AXI5-Stream only)	1	Odd parity check bit for TREADY, generated by the DUT Slave.

7. Verification Environment Structure

Root: axi_stream_master_vip_tb/

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|-- master_agent : VIP Master Agent: Sequencer, Driver (drives TVALID/TDATA/TLAST/parity), Monitor
|-- env : UVM Environment: Scoreboard, Protocol Checker, Coverage Collector, EnvConfig
|-- sequences : Stimulus library: handshake variants, sparse-stream, back-to-back, null-packet, parity-gen, parity-inject, reset, i
|-- tests : UVM test classes mapping each scenario to sequence combinations and config overrides
|-- assertions : SVA property file for real-time protocol conformance (stability, reserved-encoding, parity)
|-- top : Simulation top: DUT (Slave) instantiation, virtual interface binding, clock and reset generation
|-- verif_plan : Artifacts: YAML traceability matrix, PDF report, sub-prompt reference files

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